

Avinash Ramakrishnan Rao

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Overview

Aerospace engineer with 3+ years of industry R&D experience, pursuing an M.S. at Purdue University, specializing in advanced structural design, multiscale modeling and AI-accelerated physics simulations, seeking a Summer 2026 internship.

Education

Purdue University - MS in Aeronautics and Astronautics [3.95/4.0 GPA]

Structures, Computational Engineering | Graduate Research Assistant (GRA)

West Lafayette, IN

Aug 2024 – Dec 2026

Birla Institute of Technology - B.E. Mechanical Engineering [8.48/10 CGPA]

Undergraduate Research Student at IISc | Technical Lead SAE-eBAJA

Ranchi, India

July 2016 – May 2020

Skills

Technical: Mechanical Design; Nonlinear FEA; Multiscale Modeling; Shell/Beam Structural Modeling; ICME Methodology; Thermo-mechanical Analysis; Multidisciplinary Optimization; FEA Scripting; Computational Aerodynamics; Physics-based Surrogate Modeling; Agentic AI Workflows; API Development

Software: Catia, SolidWorks, Abaqus, Ansys, VABS, SwiftComp, FEAP, ADAMS, Gmsh, TexGen, FEniCS (In Progress)

Programming: Python, Matlab, Fortran, Mathematica, TensorFlow

Research Experience

Graduate Research Assistant, Purdue University

Graduate Advisor: Dr. Wenbin Yu

West Lafayette, IN

Aug 2024 – Present

- Established model-selection guidelines for multi-fidelity wind-turbine blade structural models with Envision Group (SSDM 2025)
- Benchmarked multi-fidelity wind-turbine blade models (beam/shell/solid), achieving **30x** faster runtime using slurm-based batch execution in Linux HPC, in collaboration with Sandia National Labs
- Developed a VABS-based thermo-mechanical beam model to predict cure-induced spring-in of L-shaped composite beams, validated against 3D FEA and experimental data
- Built an LLM-integrated, physics-based neural network pipeline for thermal-conductivity prediction of 2D woven composites, using high-fidelity multiscale simulations, accelerating design iterations by **4x** (SAMPE 2026, Journal)
- Developed an AI-driven, MSG-based multiscale modeling workflow in n8n for elastic property prediction of woven textile composites by orchestrating Gmsh, SwiftComp, and TexGen APIs

Undergraduate Research Student, Indian Institute of Science (IISc)

Advisor: Dr. Dineshkumar Harursampath

Bengaluru, India

May 2019 – Dec 2020

- Developed a zeroth-order, VAM-based geometrically exact shell element in FORTRAN (FEAP), capturing material and geometric nonlinearity while achieving **100x** faster solves than conventional shell models

Professional Experience

Mahindra Research Valley, Mahindra & Mahindra Ltd.

Senior Design Engineer - Brakes & Active Safety System

Chennai, India

Mar 2023 – Jul 2024

- Developed a methodology for nonlinear thermo-mechanical analysis of brake discs that enabled a **10%** weight reduction and **6%** coning reduction, mitigating brake judder and squeal
- Owned end-to-end Electronic Stability Control (ESC) delivery for 4 SUV programs, covering design, validation, calibration, and production line clearance including Roll and Brake Tests (RBT)
- Delivered ESC for the Australia–New Zealand market, clearing ADR-98 and meeting stringent emergency braking targets with brake locking time under **500 ms**
- Developed first-in-India off-road features including Crawl Control and Intelligent Turn Assist for the Thar Roxx model, significantly improving performance and stability on steep and low-friction surfaces

Design Engineer - Brakes & Active Safety System

Mar 2021 – Mar 2023

- Owned the complete design, analysis, and release of foundation brake components from initial CAD and FEA to validation and production for the THAR, XUV700, Scorpio and XUV300 platforms
- Resolved rig- and vehicle-level validation issues in NVH, durability, and life by reviewing test data with suppliers and updating designs, helping programs meet functional image targets and launch milestones
- Developed 1D AMESim models for brake sizing and performance mapping on ICE and EV programs, cutting early design iterations by **30%** and standardizing target deceleration curves across platforms
- Led brake component cost and warranty reduction initiatives through redesign, localization and validation, generating roughly **\$0.5 million** in annual savings while maintaining performance and reliability targets

- Led a team of 35 students in the concept, fabrication and assembly of an electric ATV for BAJA SAEINDIA, coordinating design, fabrication and testing to meet all-terrain performance targets
- Designed and manufactured suspension, steering and braking subsystems, reducing system weight by **4 kg** while maintaining durability and safety targets
- Managed sponsorships and supplier relations to complete manufacturing and testing on schedule within a **\$5,000** budget
- Represented the team through technical inspections and static/dynamic events, ensuring regulatory compliance and race readiness

Publications & Awards

Publications:

- "**Analysis of Interference-Fit Orbital Motor Using Finite Element Method in ANSYS Workbench**"
Springer Professional, Advanced Manufacturing Systems and Innovative Product Design, Feb 2021 [doi:10.1007/978-981-15-9853-1_37](https://doi.org/10.1007/978-981-15-9853-1_37)
- "**Evaluating Multi-Fidelity Structural Models for Utility-Scale Wind Turbine Blades**" SSDM Conference, June 2025.
[doi:10.1115/SSDM2025-152493](https://doi.org/10.1115/SSDM2025-152493)
- "**Thermal Conductivity Prediction of Woven Composites Using a Large Language Model-Integrated Neural Network Framework**" SAMPE Journal, Jan 2026. [doi: 10.33599/SJ.v62no1.02](https://doi.org/10.33599/SJ.v62no1.02)
- "**Thermal Conductivity Prediction of 2D Woven Composites Using an LLM-Integrated Neural Network Framework**"
ASC Conference, Oct 2025. (Proceedings forthcoming)

Awards:

- Awarded the **2025 NSF/ASC Student Travel Award** by the American Society for Composites in recognition of research excellence and to support attendance at ASC 2025 Conference
- Received the **Divisional Excellence Award** from the Global Product Development Head at Mahindra & Mahindra for resolving plant assembly and **End-of-Line (EOL)** issues, including **Roll and Brake Test (RBT)** compliance
- Received the **Divisional Excellence Award** for leading the development of **Electronic Stability Control (ESC)** for IC engine platforms
- Achieved **2nd place** nationally out of 70 teams in **Virtual BAJA SAEINDIA 2018**, recognized for innovative vehicle design, cost optimization, and strategic sales presentation
- Achieved **5th place** nationally out of 70 teams in the main **BAJA SAEINDIA 2019** competition, successfully completing all static, dynamic, and endurance events